TASK 7:- Get Basic Sales Summary from a Tiny SQLite Database using Python

sqlite_db.py - C:\Users\Avantika\OneDrive\Desktop\task7\sqlite_db.py (3.12.7)

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```python
import sqlite3

# Connect (or create) the database
conn = sqlite3.connect('sales_data.db')
cursor = conn.cursor()

# Create a simple sales table
cursor.execute('''
    CREATE TABLE IF NOT EXISTS sales (
        id INTEGER PRIMARY KEY,
        product TEXT,
        quantity INTEGER,
        price REAL
    )
''')

# Insert some sales data
sample_data = [
    ('Apple', 10, 0.5),
    ('Banana', 5, 0.3),
    ('Apple', 15, 0.5),
    ('Orange', 7, 0.7),
    ('Banana', 10, 0.3)
]
cursor.executemany('INSERT INTO sales (product, quantity, price) VALUES (?, ?, ?)', sample_data)
conn.commit()
conn.close()
```

sales_summary.py - C:\Users\Avantika\OneDrive\Desktop\task7\sales_summary.py (3.12.7)

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```python
import sqlite3
import pandas as pd

# Connect to the database
conn = sqlite3.connect('sales_data.db')

# Write SQL to get total quantity and revenue
query = '''
SELECT
    product,
    SUM(quantity) AS total_quantity,
    SUM(quantity * price) AS revenue
FROM sales
GROUP BY product
'''

# Load results into a DataFrame
df = pd.read_sql_query(query, conn)
conn.close()

# Show the summary in the console
print(df)
```

chart.py - C:\Users\Avantika\OneDrive\Desktop\task7\chart.py (3.12.7)

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```python
import sqlite3
import pandas as pd
import matplotlib.pyplot as plt

# Step 1: Connect to the SQLite database
conn = sqlite3.connect('sales_data.db')

# Step 2: SQL query to get total quantity and revenue
query = '''
SELECT
    product,
    SUM(quantity) AS total_quantity,
    SUM(quantity * price) AS revenue
FROM sales
GROUP BY product
'''

# Step 3: Load the results of the query into a pandas DataFrame
df = pd.read_sql_query(query, conn)

# Step 4: Close the database connection
conn.close()

# Step 5: Print the DataFrame (you should see the data in the console)
print(df)

# Step 6: Plot a simple bar chart
df.plot(kind='bar', x='product', y='revenue', legend=False, color='skyblue')

# Step 7: Add chart labels
plt.title('Revenue by Product')
plt.xlabel('Product')
plt.ylabel('Revenue ($)')
plt.tight_layout()

# Step 8: Show the plot
plt.show()
```

```
Type "help", "copyright", "credits" or "license()" for more information.
>>>
======== RESTART: C:\Users\Avantika\OneDrive\Desktop\task7\sqlite_db.py ========
>>>
====== RESTART: C:\Users\Avantika\OneDrive\Desktop\task7\sales_summary.py ======
  product  total_quantity  revenue
0   Apple             100     50.0
1  Banana              60     18.0
2  Orange              28     19.6
>>>
========== RESTART: C:\Users\Avantika\OneDrive\Desktop\task7\chart.py ==========
  product  total_quantity  revenue
0   Apple             100     50.0
1  Banana              60     18.0
2  Orange              28     19.6

========== RESTART: C:\Users\Avantika\OneDrive\Desktop\task7\chart.py ==========
  product  total_quantity  revenue
0   Apple             100     50.0
1  Banana              60     18.0
2  Orange              28     19.6
>>>
```

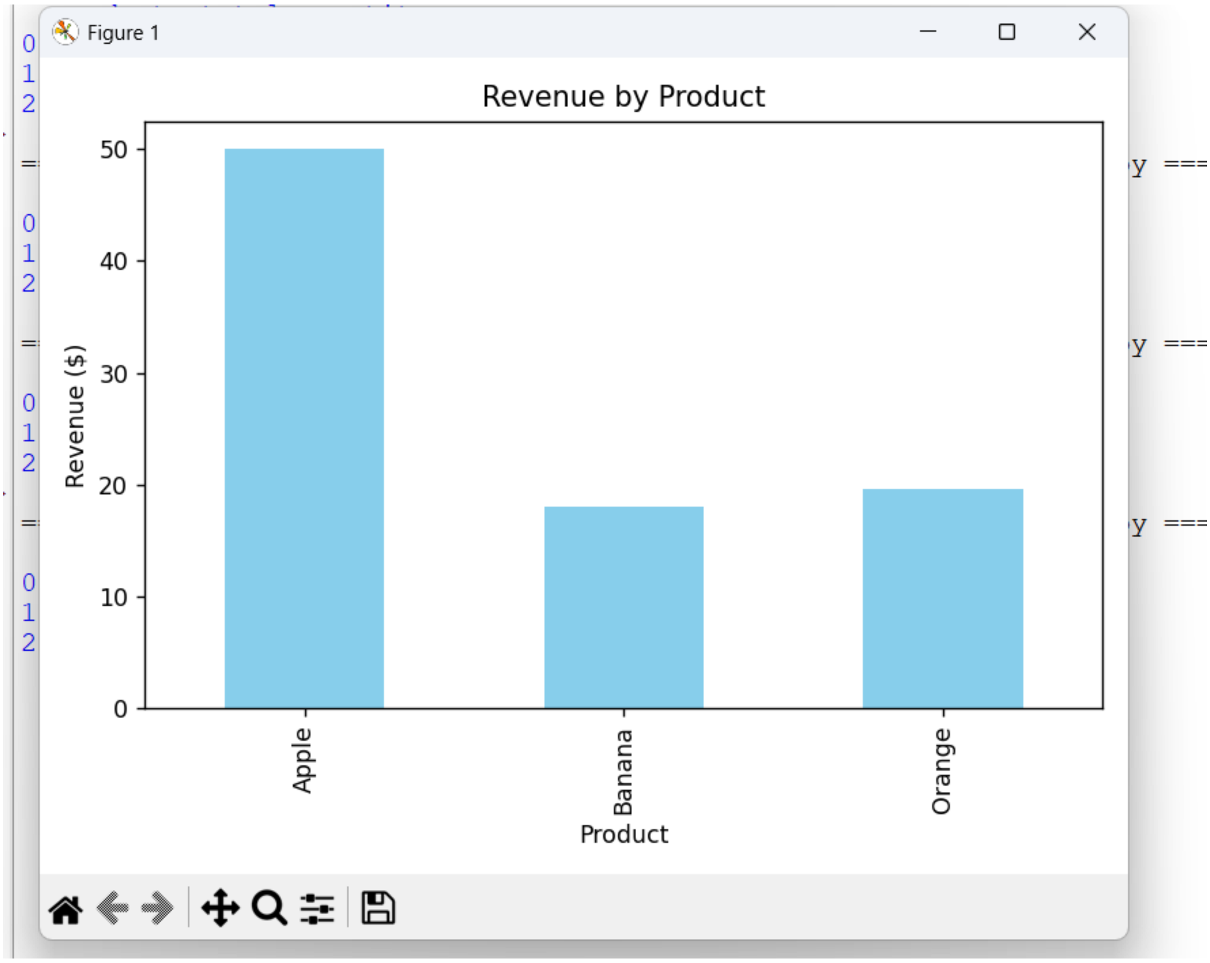
Figure 1
Revenue by Product
Revenue ($)
0
10
20
30
40
50
Apple
Banana
Orange
Product